and now it has a very efficient method of delegating workload between the CPU and the GPU.

On devices which which barely had to deal with any variety, i.e. Apple devices, the programmers can ensure a very efficient integration of the operating system with the hardware. This is why Apple's UI is very smooth even with a lot of apps installed. Windows phones are a different story, there are very few phones in the market so essentially it interface should be really smooth, however, the WP7 UI was made very graphic heavy, this mean that the GPU was now under heavy load and a characteristic lag would creep in after a few weeks of normal usage. Microsoft has fixed this with WP8. It is because of the sudden availability of extra processing power that these improvements were made possible, and also this is why your old Android, Apple or WP phone is not going to get the latest version of the OS. The hardware simply isn't enough to handle the new stuff.

How many different GPUs exist?

Since the number of operating systems are so few, the GPUs all have to be the same in its essence. This is done by ensuring that all GPUs utilise the same or at least similar instruction set. Usually manufacturers like to come up with their own instruction sets and hope they are adopted by others. This is pretty much what happened and now we have ARM Holdings Ltd., in the UK who comes out with new processor designs based on their own instruction set. This design is then licensed by other manufacturers and they modify them accordingly. Therefore, a lot of processors exist based on the most popular instruction set right now (ARMv7) which differ from each other by leaps and bounds. Then again there are companies that just license the instruction set (ARMv7) and come up with their own processor designs. The following companies are the popular players in the market:

- ARM
- DMP
- Imagem
- Imagination
- LogicBricks
- Nexus Chips
- Takumi
- TES-DST
- Think Silicon
- Vivante